

Quantitative Research Methods

Bayesian Inference

Advanced module A12 — optional self-study

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Slides, lab notebooks and data:  github.com/ChrisW09/Quantitative-Research-Methods

Where this module fits

Course chapter / module	What this module builds on it
Ch. 0 — statistics refresher	Confidence intervals and p -values — here they get rivals
Ch. 3 — linear regression	The likelihood; Bayesian regression reuses the whole fit
Ch. 4 — classification	Bayes' theorem itself (LDA already used it for labels)
Ch. 6 — regularisation	Ridge <i>is</i> a Bayesian estimate in disguise — proved here

The other way to be uncertain

The whole taught course is frequentist: parameters are fixed, data are random. This module flips it — **treat the parameter as the random thing**, update beliefs with data, and read off statements the frequentist machinery cannot make: “the probability that p exceeds 0.25 is 0.93.”

One of twelve optional advanced modules; self-study. Every number on these slides is reproduced by `make_figures.py` in the module folder.

Contents

- 1 The Problem
- 2 The Beta-Binomial Engine
- 3 Gaussian Means & Shrinkage
- 4 Computation: MCMC
- 5 Regression & Decisions
- 6 Python Lab
- 7 Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this module

Symbol	Meaning
θ	the unknown parameter, now carrying a distribution
$\pi(\theta)$	the prior: belief before the data
$L(\theta) = p(\text{data} \mid \theta)$	the likelihood (Chapters 3–4's object)
$\pi(\theta \mid \text{data})$	the posterior: belief after the data
$\text{Beta}(a, b)$	the workhorse prior for a probability
CrI	credible interval (posterior probability interval)
MAP	maximum a posteriori: the posterior's peak
τ	prior standard deviation (how sure you are beforehand)

Sources and references

Primary textbook

James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning, with Applications in Python*, Springer. — Supplies the likelihoods, ridge, and the datasets (Advertising.csv).

Canonical further reading (optional)

Gelman et al., *Bayesian Data Analysis* (3rd ed., 2013); McElreath, *Statistical Rethinking* (2nd ed., 2020) — the standard references, one formal, one narrative.

Learning objectives

By the end of this module you will be able to:

- **State** Bayes' theorem for parameters and the roles of prior, likelihood and posterior.
- **Update** a Beta prior with binomial data by hand, and report a posterior mean and credible interval.
- **Explain** every posterior mean in this module as a *precision-weighted compromise* between prior and data.
- **Derive** ridge regression as the MAP estimate under a Gaussian prior — and translate $\lambda \leftrightarrow \tau$.
- **Implement** a Metropolis sampler from scratch and check it against the analytic answer.
- **Distinguish** credible from confidence intervals without garbling either.

Roadmap of this module

Believe, observe, update

- 1 The problem: what kind of probability a parameter can have.
- 2 The Beta-binomial engine: updating by hand, priors, credible intervals.
- 3 Gaussian means and shrinkage: precision weighting — and ridge unmasked.
- 4 Computation: when conjugacy runs out, walk (Metropolis).
- 5 Bayesian regression and decisions: Advertising, and an A/B call.
- 6 Python lab, summary, appendix.

Outline

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Two questions that sound the same

Frequentist (Ch. 0)	Bayesian (this module)
How would my estimate vary across imaginary repeated samples?	What should I believe about θ , given the one sample I have?
Parameter fixed, data random	Data fixed, parameter random
p -value, confidence interval	posterior, credible interval
"95% of such intervals cover"	"95% probability θ is in here"

Rule (Bayes' theorem for parameters)

$$\pi(\theta \mid \text{data}) \propto L(\theta) \pi(\theta)$$

$$\text{posterior} \propto \text{likelihood} \times \text{prior.}$$

How to read this

Chapter 4's LDA already ran Bayes' theorem — on *class labels*. The only new move is to run it on *parameters*. The likelihood is the same object Chapters 3–4 maximised; the prior is new; the posterior is the answer.

Exercise A12.1 — Credible or confident? [Concept]

Exercise

A 95% *confidence* interval for a conversion rate is $[0.19, 0.31]$; a 95% *credible* interval from the same data is $[0.19, 0.31]$ as well.

- 1 Write the one-sentence claim each interval is entitled to make.
- 2 A manager says: “there is a 95% chance the true rate is between 0.19 and 0.31.” Which interval licenses that sentence?
- 3 Why do the two intervals often nearly coincide anyway?

Solution — Exercise A12.1

Solution

Part 1. Confidence: “this interval was built by a recipe that captures the true rate in 95% of repeated samples” — a statement about the *procedure*. Credible: “given the data and the prior, the posterior probability that the rate lies in $[0.19, 0.31]$ is 0.95” — a statement about *this* interval.

Part 2. Only the credible interval. The manager’s sentence assigns a probability to the parameter, which frequentist logic forbids: once computed, a confidence interval either contains the fixed truth or does not. (Everyone reads confidence intervals the manager’s way — Chapter 0 flagged the misreading; here it becomes a licensed statement.)

Part 3. With a flat-ish prior and decent n , the posterior is essentially the normalised likelihood, so both intervals track the same curvature — they agree numerically while meaning different things. The divergence appears when priors are informative or n is small.

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Conjugacy: the update you can do in your head

Rule (Beta-binomial conjugacy)

prior $\text{Beta}(a, b)$ + y successes in n trials $\implies$ posterior $\text{Beta}(a + y, b + n - y)$

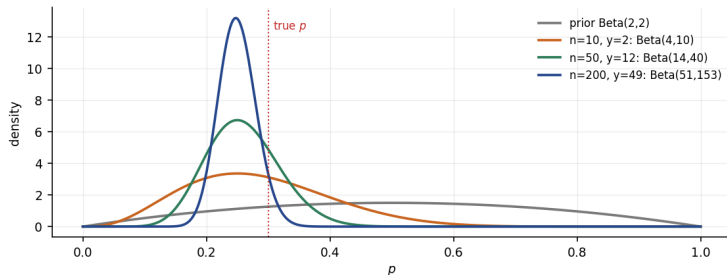
How to read this

The prior acts like $a + b$ *imaginary trials*, a of them successes. Updating is bookkeeping: add real successes to a , real failures to b . Posterior mean $= \frac{a+y}{a+b+n}$ — always *between* the prior mean and the sample proportion.

A conversion rate, true $p = 0.3$

Prior $\text{Beta}(2, 2)$. After $n = 10$ ($y = 2$): $\text{Beta}(4, 10)$, mean 0.286. After $n = 50$ ($y = 12$): $\text{Beta}(14, 40)$, mean 0.259. After $n = 200$ ($y = 49$): $\text{Beta}(51, 153)$, mean 0.250, 95% CrI [0.193, 0.312].

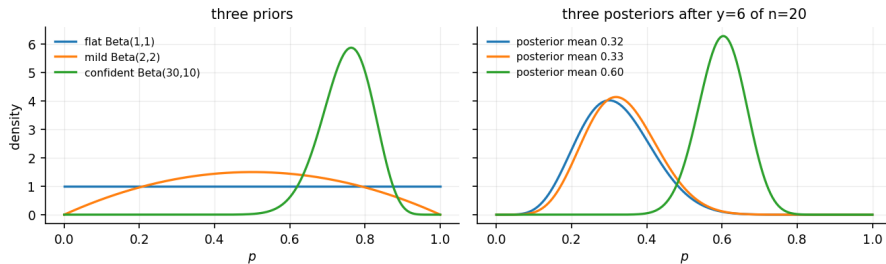
Watching belief sharpen



Data overwhelm the prior — at the data's own pace

The gentle $\text{Beta}(2,2)$ hump narrows steadily; by $n = 200$ the posterior is tight around the sample rate and the prior's four imaginary trials are outvoted 50 : 1. The truth (0.3) sits inside every credible interval along the way.

Three priors, one dataset



Priors matter exactly as much as they claim to

Same data ($y = 6$ of $n = 20$). Flat prior $\rightarrow$ mean 0.318; mild Beta(2, 2) $\rightarrow$ 0.333; a confident-but-wrong Beta(30, 10) (forty imaginary trials at 75%) $\rightarrow$ 0.600 — the data are outvoted 2:1. A prior is a claim with a stake; make it inspectable, and show the sensitivity.

Exercise A12.2 — One update by hand [Math]

Exercise

Prior $\text{Beta}(2, 2)$ on a click-through rate; you observe 7 clicks in 10 impressions.

- 1 Write down the posterior distribution.
- 2 Compute the prior mean, the MLE, and the posterior mean.
- 3 Show that the posterior mean is a weighted average $w \cdot \text{MLE} + (1 - w) \cdot \text{prior mean}$, and identify w .

Solution — Exercise A12.2

Solution

Part 1. $\text{Beta}(2 + 7, 2 + 3) = \text{Beta}(9, 5)$.

Part 2. Prior mean = $2/4 = 0.5$; MLE = $7/10 = 0.7$; posterior mean = $9/14 \approx 0.643$.

Part 3. With $a + b = 4$ prior trials and $n = 10$ real ones,

$$\frac{a + y}{a + b + n} = \underbrace{\frac{n}{n + a + b}}_{w=10/14} \cdot \frac{y}{n} + \underbrace{\frac{a + b}{n + a + b}}_{1-w=4/14} \cdot \frac{a}{a + b} = \frac{10}{14}(0.7) + \frac{4}{14}(0.5) = 0.643.$$

The weight on the data is *sample size against imaginary sample size* — the entire prior-vs-data politics of this module in one fraction.

Common mistake

Reporting the MAP ($8/12 = 0.667$) as “the Bayesian estimate” without saying so. Mean, median and MAP all summarise the posterior; for skewed posteriors they differ, and the honest report is the whole interval.

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The Gaussian mean: precision-weighted compromise

Rule (known σ , prior $\mu \sim N(\mu_0, \tau^2)$)

$$\text{posterior mean} = w \bar{x} + (1 - w) \mu_0, \quad w = \frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2}$$

posterior variance = $(n/\sigma^2 + 1/\tau^2)^{-1}$.

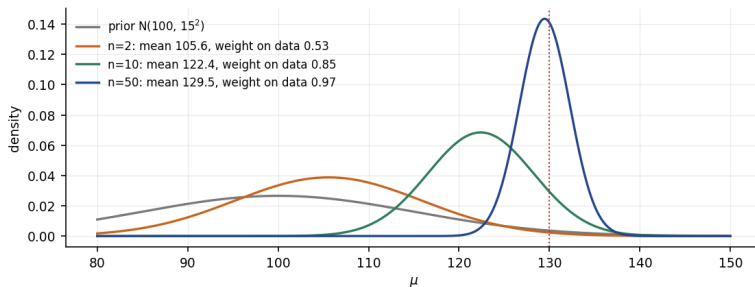
How to read this

Precisions (inverse variances) *add*; means combine *weighted by precision*. The data's vote grows linearly in n ; the prior's vote is fixed — the Beta-binomial story again, in Gaussian dress.

Prior $N(100, 15^2)$, $\sigma = 20$, truth 130

$n = 2$: mean 105.6, weight on data 0.53. $n = 10$: 122.4, weight 0.85. $n = 50$: 129.5, weight 0.97 — the prior fades on schedule.

The prior fading, drawn



Shrinkage is not a bug

At $n = 2$ the posterior barely moves off the prior — and that is the *correct* behaviour: two noisy observations should not overturn a considered belief. Every shrinkage estimator in Chapter 6 is this picture wearing an optimisation costume, as the next slide proves.

Ridge regression, unmasked

Claim

Ridge (Ch. 6) is the **MAP estimate** of a linear model under the prior $\beta_j \sim N(0, \tau^2)$, with $\lambda = \sigma^2 / \tau^2$.

The two-line proof

Maximising the log-posterior = log-likelihood + log-prior:

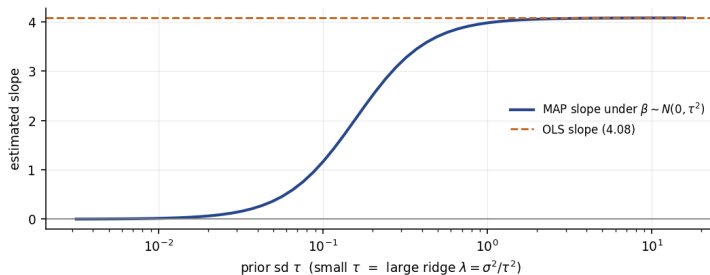
$$-\frac{1}{2\sigma^2} \sum_i (y_i - x_i^\top \beta)^2 - \frac{1}{2\tau^2} \sum_j \beta_j^2 \propto -\left[\sum_i (y_i - x_i^\top \beta)^2 + \frac{\sigma^2}{\tau^2} \sum_j \beta_j^2 \right],$$

which is exactly the ridge criterion with $\lambda = \sigma^2 / \tau^2$.

Chapter 6's tuning dial, translated

A large penalty λ is a tight prior (τ small). Cross-validating λ empirically chooses how strong a prior the data can support. (The lasso? A Laplace prior — same construction.)

The dictionary, drawn on Advertising



Read the dial

Standardised sales-on-TV slope: OLS says 4.08. As the prior tightens ($\tau \downarrow$), the MAP slope shrinks toward 0 — at $\tau = 0.1$ (i.e. $\lambda = 500$) it is 1.16. Chapter 6's whole coefficient-path figure is this curve, one predictor at a time.

Exercise A12.3 — Precision weighting by hand [Math]

Exercise

Quality control believes a machine fills bottles at $\mu_0 = 100\text{ml}$ ($\tau = 15$). Measurement noise is $\sigma = 20$. A morning sample gives $n = 4$, $\bar{x} = 120$.

- 1 Compute the posterior mean and standard deviation of μ .
- 2 The engineer says “the sample says 120, so recalibrate to 120.” What does the posterior say instead, and why?
- 3 How large would n (at $\bar{x} = 120$) have to be for the weight on the data to reach 0.9?

Solution — Exercise A12.3

Solution

Part 1. Precisions: prior $1/\tau^2 = 1/225 \approx 0.00444$; data $n/\sigma^2 = 4/400 = 0.01$. Weight on data $w = 0.01/0.01444 \approx 0.692$. Posterior mean $= 0.692 \times 120 + 0.308 \times 100 \approx 113.8$; posterior sd $= \sqrt{1/0.01444} \approx 8.3$.

Part 2. Split the difference toward 113.8: with only four noisy measurements ($SE = 20/2 = 10$), a 20ml jump from a well-calibrated prior is as likely to be sampling noise as drift. The posterior *is* the rational compromise — and its sd (8.3) says a follow-up sample is worth taking before touching the machine.

Part 3. Need $\frac{n/400}{n/400 + 1/225} = 0.9 \Rightarrow n/400 = 9/225 \Rightarrow n = 16$.

Common mistake

Adding standard deviations instead of precisions. Only *inverse variances* add; the posterior sd is always *smaller* than both the prior's and the data's alone.

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When conjugacy runs out, walk

Real models rarely have closed-form posteriors. The general tool: build a random walk whose long-run visiting frequencies *are* the posterior.

Algorithm (Metropolis)

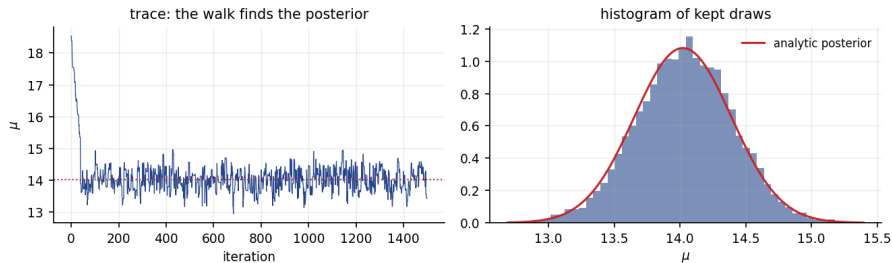
From current θ : (1) propose $\theta' = \theta + N(0, s^2)$; (2) accept with probability $\min(1, \pi(\theta' | \text{data})/\pi(\theta | \text{data}))$, else stay; (3) record, repeat. Discard the warm-up (“burn-in”), keep the rest.

Why it works, in one sentence

Moves uphill are always taken and moves downhill sometimes — calibrated so the walk spends time in each region *in proportion to its posterior probability*; only posterior *ratios* are needed, so the intractable normalising constant cancels.

Like A2's Shapley values and A3's conformal intervals, we build it in 15 lines; the production tools are PyMC, Stan, NumPyro.

Metropolis on real data: the mean of Advertising sales



Sampler vs analytic truth

20,000 steps, acceptance rate 0.62, burn-in 2,000. Posterior mean 14.04 vs analytic 14.02; posterior sd 0.367 vs analytic 0.369. The trace (left) starts at a deliberately wrong value and finds the posterior within a few hundred steps.

Exercise A12.4 — Reading a sampler [Concept]

Exercise

Three colleagues run Metropolis on the same posterior and show you their diagnostics:

- 1 Colleague A: acceptance rate 0.99; the trace drifts in slow waves.
- 2 Colleague B: acceptance rate 0.02; the trace is long flat plateaus.
- 3 Colleague C: acceptance rate 0.55; the trace is a fuzzy caterpillar.

Diagnose each, name the shared root cause for A and B, and state the fix.

Solution — Exercise A12.4

Solution

A — steps too small. Nearly every proposal is accepted because it barely moves; the chain explores the posterior like a snail, and consecutive draws are almost copies. Estimates converge, but painfully slowly.

B — steps too large. Proposals leap into low-probability territory and are almost always rejected; the chain freezes for hundreds of iterations. Same disease as A — *high autocorrelation in the chain* — reached from the opposite direction.

C — healthy. Mid-range acceptance with fast mixing; rules of thumb put the sweet spot around 0.2–0.5 for random-walk Metropolis (our run: 0.62, acceptably brisk for one dimension).

Fix. Tune the proposal scale s toward the sweet spot; then confirm with the trace and by running a second chain from a different start and checking both agree.

Common mistake

Judging a sampler by acceptance rate alone. It is a symptom, not the disease: the disease is autocorrelation, and the trace plot (plus multiple chains) is the diagnostic that sees it.

Extended Exercise A12.1 — Your own sampler [Python]

Extended exercise (15 min)

Implement Metropolis from scratch for the mean of Advertising sales (treat σ as the sample sd; prior $\mu \sim N(0, 100^2)$):

- 1 Write the log-posterior and the accept/reject loop; run 20,000 steps at proposal scale 0.5 from the (bad) start $\bar{x} + 5$.
- 2 Report acceptance rate, posterior mean and sd after burn-in; compare with the analytic values (14.02, 0.369).
- 3 Re-run with proposal scales 0.01 and 50. Connect what you see to Exercise A12.4.

Solution — Extended Exercise A12.1 (1/2) — code

```

import numpy as np, pandas as pd
x = pd.read_csv("Advertising.csv", index_col=0)["sales"].to_numpy()
rng = np.random.default_rng(2024)
sigma, tau, B, s = x.std(ddof=1), 100.0, 20_000, 0.5

def logpost(mu): # log posterior, up to a constant
    return -0.5*((x - mu)**2).sum()/sigma**2 - 0.5*mu**2/tau**2

chain, cur = np.empty(B), x.mean() + 5
lp = logpost(cur); acc = 0
for i in range(B):
    prop = cur + rng.normal(0, s)
    lp_prop = logpost(prop)
    if np.log(rng.uniform()) < lp_prop - lp: # uphill: always; downhill: sometimes
        cur, lp, acc = prop, lp_prop, acc + 1
    chain[i] = cur
kept = chain[2000:]
print(f"accept {acc/B:.2f}, mean {kept.mean():.2f}, sd {kept.std(ddof=1):.3f}")

```

Solution — Extended Exercise A12.1 (2/2)

Solution — parts 2 and 3

Part 2. Acceptance ≈ 0.62 ; posterior mean 14.04, sd 0.367 — against analytic 14.02 and 0.369. A dozen lines of NumPy reproduce the exact answer to two decimals; that agreement *is* the test that the sampler is correct.

Part 3. At $s = 0.01$ the acceptance rate rockets toward 1 and the trace crawls (colleague A); at $s = 50$ acceptance collapses toward 0 and the trace is plateaus (colleague B). Both chains still *eventually* target the right posterior — the theory guarantees the destination, tuning buys the arrival time.

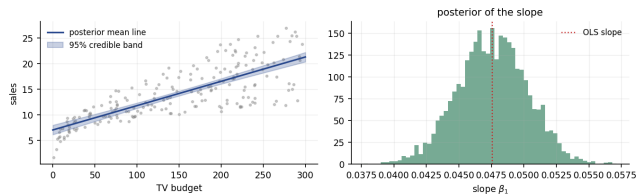
Common mistake

Using $\pi(\theta')/\pi(\theta)$ on the raw (non-log) scale. With $n = 200$ the likelihood underflows `float64` long before the ratio is formed — always accept/reject on the *log* scale, as the code does.

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Bayesian simple regression: sales on TV



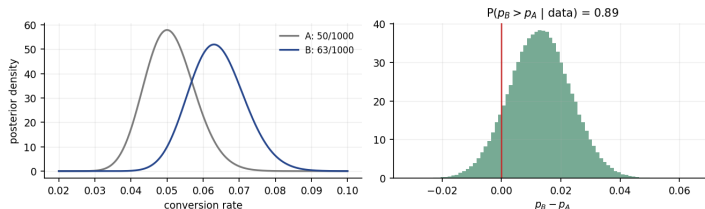
The posterior of a Chapter 3 slope

Weak $N(0, 10^2)$ priors on both coefficients. Posterior slope: mean 0.0476, sd 0.0027 — numerically the OLS fit (0.0475) with its standard error, *reinterpreted*: the left panel's band is a 95% credible band of regression lines drawn from the posterior.

Weak priors reproduce Chapter 3 — by design

With flat-ish priors, Bayesian regression rediscovers least squares; the machinery pays when priors carry information — small n , correlated predictors (ridge!), hierarchies of related groups.

A decision, not a p -value: the Bayesian A/B test



A: 50/1000, B: 63/1000

Flat priors, two independent Beta posteriors, 200,000 paired draws: $\Pr(p_B > p_A | \text{data}) = 0.895$ — a directly decision-shaped number. (The two-sided frequentist test at these counts does not clear 0.05; both summaries are honest, they answer different questions.)

Common mistake

Believing posteriors are immune to module A1's peeking problem: if you *stop when the posterior looks good*, the long-run error rate of that *policy* still inflates — design the stopping rule, whatever your school.

Extended Exercise A12.2 — The Bayesian A/B memo [Integrative]

Extended exercise (15 min)

Marketing ran the experiment above (A: 50/1000, B: 63/1000) and asks for a recommendation memo.

- 1 Reproduce $\Pr(p_B > p_A \mid \text{data})$ by simulation with flat priors. State the number with its meaning in one sentence.
- 2 Compute the posterior of the *lift* $p_B - p_A$ and report a 95% credible interval. Does it contain 0, and does that contradict part 1?
- 3 Switching costs EUR 2,000; each conversion is worth EUR 5 on 100,000 annual visitors. Turn the posterior into an expected-value decision.
- 4 State one sentence of prior sensitivity: what would a sceptical prior (e.g. Beta(50, 950) on both arms) do to the conclusion?

Solution — Extended Exercise A12.2

Solution

Part 1. Draw from $\text{Beta}(51, 951)$ and $\text{Beta}(64, 938)$, compare: $\Pr(p_B > p_A) \approx 0.895$ — “given the data, B is better with probability about 0.9.”

Part 2. The lift’s posterior is centred near 0.013 with 95% CrI roughly $[-0.007, 0.033]$ — it *does* straddle 0. No contradiction: the CrI asks “what values are plausible”, part 1 asks “which arm is more likely better”; a 0.895 direction is compatible with an interval touching zero.

Part 3. Expected lift $\approx 0.013 \times 100,000 \times \text{EUR } 5 = \text{EUR } 6,500/\text{year}$ against EUR 2,000 once; even weighting the downside draws, expected value favours switching — recommend B, note the $\sim 10\%$ chance it is not better, and (module A1) confirm with a pre-registered follow-up if the cost were larger.

Part 4. A sceptical $\text{Beta}(50, 950)$ prior (five percent, held with 1000 trials’ conviction) roughly halves the estimated lift and pulls $\Pr(p_B > p_A)$ toward 0.8 — the direction survives, the enthusiasm moderates. Report both runs.

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Python lab — run it live

Reproduce every number: `make_figures.py`

Every number on these slides is computed by `make_figures.py` in this module's folder, from the bundled `Advertising.csv` and seeded simulations — run it and read it alongside the deck:

- the Beta-binomial updates ($\text{Beta}(2, 2) \rightarrow \text{Beta}(51, 153)$) and the three-priors picture;
- the precision-weighted Gaussian posteriors ($w = 0.53 \rightarrow 0.97$);
- the from-scratch Metropolis run (14.04 vs analytic 14.02);
- Bayesian regression on `Advertising` and the ridge-MAP dial;
- the A/B posterior ($\Pr(p_B > p_A) = 0.895$).

A companion notebook in the course lab format is in preparation; until it lands, the script *is* the lab — it runs top to bottom in the course environment with no extra packages.

The whole module in twelve lines

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(2024)

# belief -> data -> updated belief, all in one conjugate line
a, b = 2, 2 # prior Beta(2,2): four imaginary trials
y, n = 49, 200 # the data
post = stats.beta(a + y, b + n - y)
print(f"posterior mean {post.mean():.3f}, "
      f"95% CrI [{post.ppf(0.025):.3f}, {post.ppf(0.975):.3f}]" )

# and the sentence the frequentist interval cannot say:
print(f"P(p > 0.25 | data) = {1 - post.cdf(0.25):.2f}")
```

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Module A12 in one slide

What we accomplished

A second, coherent grammar for uncertainty — run on the course's own likelihoods and datasets.

- Posterior $\propto$ likelihood $\times$ prior — Chapter 4's theorem, aimed at parameters.
- Beta-binomial updating by hand: Beta(2, 2) plus 200 trials $\rightarrow$ Beta(51, 153), CrI [0.193, 0.312].
- Every posterior mean is a precision-weighted compromise; the data's weight grows as $n/(n + \text{imaginary } n)$.
- Ridge unmasked: MAP under $N(0, \tau^2)$ priors, $\lambda = \sigma^2/\tau^2$ — Chapter 6 was Bayesian all along.
- Metropolis from scratch: 14.04 ± 0.367 against analytic 14.02 ± 0.369 .
- Decision-shaped output: $\Pr(p_B > p_A \mid \text{data}) = 0.895$ — with the peeking caveat imported from A1.

Key formulas at a glance

Bayes for parameters $\pi(\theta \mid \text{data}) \propto L(\theta) \pi(\theta)$.

Beta-binomial $\text{Beta}(a, b) + (y, n) \rightarrow \text{Beta}(a+y, b+n-y)$; mean $(a+y)/(a+b+n)$.

Shrinkage weight posterior mean $= w \hat{\theta}_{\text{data}} + (1-w) \theta_{\text{prior}}$, $w = n/(n+a+b)$ (Beta) or $\frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2}$ (Gaussian).

Gaussian posterior precision $= n/\sigma^2 + 1/\tau^2$; precisions add, never sds.

Ridge as MAP $\hat{\beta}_{\text{ridge}} = \arg \max$ posterior under $\beta_j \sim N(0, \tau^2)$, with $\lambda = \sigma^2/\tau^2$.

Metropolis accept θ' w.p. $\min\{1, \pi(\theta' \mid \text{data})/\pi(\theta \mid \text{data})\}$ — ratios only, log scale in code.

Vocabulary checklist

Term	Meaning
prior	belief about θ before the data, as a distribution
likelihood	$p(\text{data} \mid \theta)$ — Chapters 3–4's fit criterion
posterior	updated belief; the module's universal output
conjugate prior	family the posterior stays inside (Beta for binomial)
credible interval	interval holding stated posterior probability
MAP	posterior mode; ridge and lasso are MAPs
shrinkage	posterior pulled from data toward prior
burn-in	discarded warm-up of an MCMC chain
acceptance rate	share of Metropolis proposals taken; tune toward mid-range
prior sensitivity	re-running with other priors; always reported

Decision rules of thumb

- Small n , real prior knowledge, or a decision to make $\Rightarrow$ Bayesian machinery earns its keep. Large n , flat priors $\Rightarrow$ it will agree with Chapter 0; use whichever is cheaper.
- Choose priors you can defend out loud, in units of imaginary observations — and always show a sensitivity run.
- Report the whole posterior (or its CrI), not just a point; say *which* point if you must give one.
- Conjugate first, grid second, MCMC when you must; always check a sampler against a case with a known answer.
- Trace plot + second chain before trusting any MCMC number.
- Probability statements about parameters: credible intervals only — never launder a confidence interval into one.

Common pitfalls

Don't

- 1 Read a confidence interval as “95% probability the parameter is inside” — that sentence needs a posterior.
- 2 Hide an informative prior (Beta(30, 10) moved the answer from 0.32 to 0.60 on the same data).
- 3 Add standard deviations where only precisions add.
- 4 Compute likelihood ratios off the log scale ($n = 200$ already underflows).
- 5 Trust a chain by its acceptance rate alone — plateaus and snails both lie.
- 6 Peek-and-stop on a posterior and claim the error rate of a fixed- n design.

Self-check questions

Quick self-assessment

- ❶ Prior $\text{Beta}(2, 2)$, then 7 successes in 10 trials: posterior, posterior mean, and the weight on the data?
- ❷ Prior $N(100, 15^2)$, $\sigma = 20$, $n = 4$, $\bar{x} = 120$: posterior mean?
- ❸ State the ridge–Bayes dictionary in one equation.
- ❹ Your Metropolis run has acceptance 0.99 and a slowly drifting trace. Diagnosis and fix?
- ❺ $\Pr(p_B > p_A \mid \text{data}) = 0.895$ while the 95% CrI for $p_B - p_A$ includes 0. Contradiction?
- ❻ What is the one sentence a credible interval may say that a confidence interval may not?

Answers

(1) $\text{Beta}(9, 5)$; $9/14 \approx 0.643$; $w = 10/14 \approx 0.71$. (2) $w \approx 0.69$, mean ≈ 113.8 . (3) $\lambda = \sigma^2/\tau^2$ under $\beta_j \sim N(0, \tau^2)$. (4) Proposal scale too small — enlarge s toward acceptance ~ 0.2 – 0.5 , confirm on the trace. (5) No: one is a statement about direction, the other about plausible magnitudes; both are read off the same posterior. (6) “Given the data, the parameter lies in this interval with probability 0.95.”

Five things to remember

Module A12 in five lines

- 1 **Posterior $\propto$ likelihood $\times$ prior** — one line, the whole framework.
- 2 **Every estimate is a compromise weighted by precision** — the data's weight is n against the prior's imaginary n .
- 3 **Ridge is a prior in disguise: $\lambda = \sigma^2/\tau^2$** — Chapter 6 has been Bayesian all along.
- 4 **MCMC trades formulas for a walk** — and a from-scratch Metropolis matched the analytic answer to two decimals.
- 5 **Priors are claims with stakes**: state them, price them in imaginary trials, and show the sensitivity run.

What's next

- **Module A1 — Randomised Controlled Trials:** the A/B decision slide is A1's experiment wearing posterior clothes — and A1's peeking warning survives the change of school.
- **Chapter 6, re-read:** the ridge and lasso paths are prior strength dialled from ∞ to 0; the Bayesian reading explains *why* shrinking helps when p is large.
- **Module A3 — Conformal Prediction:** the frequentist route to calibrated uncertainty — guarantees without priors, at the price of answering only about *predictions*. Reading A3 and A12 together maps the whole uncertainty landscape.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **The Beta-binomial update, derived** — two lines of algebra; moved out because the recipe, not the proof, carries the deck.
- **A conjugate-pairs reference table** — look-up material.
- **The posterior predictive** — forecasting a new observation, not a parameter; the bridge to Chapters 3 and A3.

The Beta–binomial update, derived

Prior density $\pi(p) \propto p^{a-1}(1-p)^{b-1}$; binomial likelihood $L(p) \propto p^y(1-p)^{n-y}$. Multiply:

$$\pi(p | y) \propto p^y(1-p)^{n-y} \cdot p^{a-1}(1-p)^{b-1} = p^{(a+y)-1}(1-p)^{(b+n-y)-1},$$

which is the kernel of $\text{Beta}(a+y, b+n-y)$ — no integral was ever computed; recognising the *shape* suffices.

Why conjugacy is a big deal

The prior and posterior live in the same family, so updating is arithmetic on two counters (a, b) — streaming data update beliefs in $O(1)$, and the “imaginary trials” reading of the prior is literal.

Conjugate pairs at a glance

Likelihood	Conjugate prior	Posterior update
Binomial (n, p)	Beta(a, b)	$a+y, b+n-y$
Poisson (λ)	Gamma(a, b)	$a + \sum y_i, b + n$
Gaussian mean (known σ)	$N(\mu_0, \tau^2)$	precision-weighted (main deck)
Gaussian variance (known μ)	Inverse-Gamma	shape + $n/2$
Multinomial	Dirichlet	add the category counts

How to use this table

Each row is the same story: the prior is phrased as pseudo-data, and updating adds real data to it. Module A4's exponential-family machinery is the deep reason this table exists: every exponential-family likelihood has a conjugate family of exactly this form.

The posterior predictive: betting on the next observation

For a new observation $\tilde{y}$, integrate predictions over the posterior:

$$p(\tilde{y} \mid \text{data}) = \int p(\tilde{y} \mid \theta) \pi(\theta \mid \text{data}) d\theta.$$

The next conversion

After Beta(51, 153): $\Pr(\tilde{y} = 1 \mid \text{data}) = E[p \mid \text{data}] = 51/204 = 0.25$ — but for the next *twenty* visitors the predictive is Beta-binomial, *wider* than Binomial(20, 0.25), because parameter uncertainty rides along.

The bridge to prediction intervals

Chapter 3's prediction intervals and module A3's conformal sets solve this same problem in frequentist dress: honest forecasts must carry *both* noise and parameter uncertainty. The posterior predictive is the Bayesian statement of that requirement.